

AI Ethics

from a computer scientist's perspective

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Agency

What is Agency?

Agency

The capacity to make autonomous choices and take intentional actions that affect oneself and the world. Agency requires consciousness, intentionality, and the ability to understand consequences of one's actions.

Core characteristics of agency:

Autonomous decision-making based on goals and values

Intentionality and purposeful action

Understanding and evaluating consequences

Moral responsibility for outcomes

Human Agency vs. AI Systems

Human Agency

Possesses: Consciousness, intentionality, understanding of context and meaning

Creates: Goals, values, and purposes based on lived experience

Bears: Moral responsibility and accountability for decisions and their consequences

Can: Reflect on actions, adapt reasoning, and make ethical judgments

AI Systems (Current)

Lacks: Consciousness, genuine understanding, or subjective experience

Executes: Patterns learned from data without comprehension of meaning or purpose

Cannot: Bear moral responsibility—responsibility remains with human designers and deployers

Appears: Agentic through sophisticated pattern matching, not true agency

Why Agency Matters for AI Ethics

Agency is the organizing principle for applying ethical frameworks to AI systems

1. Responsibility Assignment

Since (current) AI lacks agency, ethical responsibility cannot be delegated to algorithms. Computer scientists and organizations deploying AI must take ownership of all outcomes and impacts.

2. Design as Ethical Act

Every technical decision, from data collection to model architecture to deployment context, embeds ethical choices. These are human decisions with moral weight.

3. Focus on Human Agency

Ethics in AI is fundamentally about how human agency designs, deploys, and governs systems that affect other humans. The question is (currently) not "is the AI ethical?" but "are we being ethical in how we create and use AI?"

The Agency Framework

Three Core Questions for AI Ethics

1

Who has agency?

- Humans (designers, companies, users, regulators)
- AI systems (appear to have agency, but do they really?)
- Both? Neither? (responsibility gaps)

2

Who are the moral subjects versus objects?

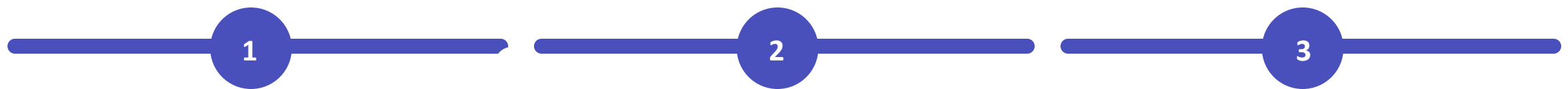
- Subjects (agents with moral responsibility)
- Objects (those affected)
- Confused category (AI treated as subject but is it?)

3

Who should have agency?

- Where should humans maintain control?
- Where can we safely delegate to AI?
- How do we preserve human agency?

Ethics for...?



1

Designing AI systems

- agency with engineers and other stakeholders

2

AI itself – how it should behave

- initially, as AI designers designed it
- agency with AI?

3

Using AI

- agency with you (+ possibly with AI)

Philosophical Frameworks

Three Ethical Frameworks

When you have agency, how do you reason about ethical decisions? Here are three major philosophical frameworks that provide different lenses for ethical reasoning:

Consequentialism

Core principle: Actions are judged by their outcomes—maximize good, minimize harm

Deontology

Core principle: Rules, duties, and rights matter regardless of outcomes—some things are simply wrong

Virtue Ethics

Core principle: Character and intentions matter—what kind of person are you being?

Virtue Ethics

Core Philosophy

Principle: Character and intentions matter. Focus on virtues, not rules or outcomes

- What kind of person should I be?
- Cultivate virtues: honesty, courage, wisdom, justice
- Character developed through practice

Key thinker: Aristotle

For AI Designers

What kind of culture and character does your design work cultivate?
What virtues guide your decisions?

Example: Building addictive features cultivates greed; accessibility cultivates justice

Strengths: Emphasizes moral development and culture

Limitations: Less clear guidance for specific decisions

Deontology (Kant)

Core Philosophy

Principle: Rules and duties matter regardless of outcomes

- Some actions are wrong in themselves
- Respect for human dignity and autonomy
- Treat people as ends, never merely as means

Key thinker: Immanuel Kant

Categorical Imperative: Act only according to maxims you'd want to be universal law

For AI Designers

Some practices are wrong even if beneficial: privacy violations, manipulation, treating users as mere means

Example: Dark patterns that manipulate users are wrong regardless of business outcomes

Strengths: Protects fundamental rights, clear boundaries, human dignity central

Limitations: Rules can conflict; rigid; doesn't help prioritize when rights clash

Consequentialism / Utilitarianism

Core Philosophy

Principle: Actions are judged entirely by their outcomes

- Maximize aggregate good
- Minimize aggregate harm
- "Greatest good for greatest number"

Key thinkers: Jeremy Bentham, John Stuart Mill

Central question: What produces the best overall consequences?

For AI Designers

Weigh total welfare impact of deploying systems: Who benefits, who's harmed, by how much?

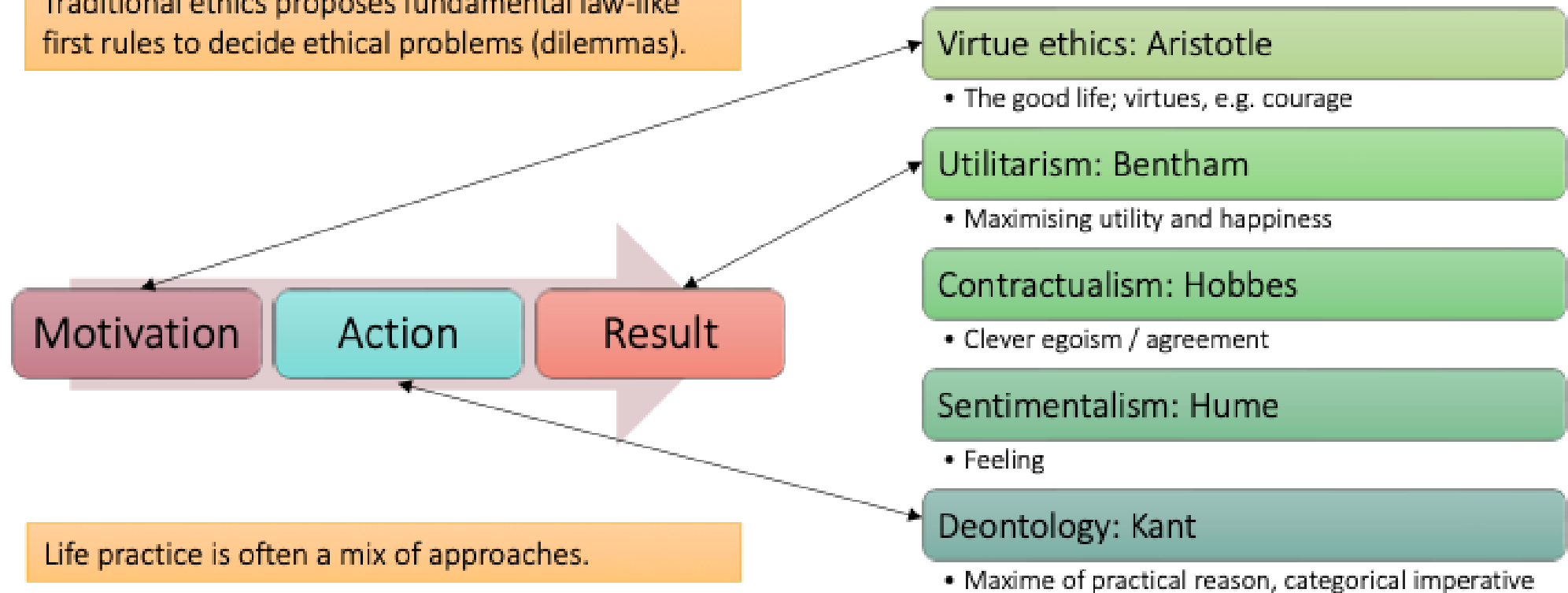
Example: Deploy facial recognition if aggregate benefits outweigh aggregate harms?

Strengths: Practical, measurable, policy-relevant

Limitations: Hard to predict consequences; can justify harming minorities for majority benefit; whose welfare counts?

Types of ethics

Traditional ethics proposes fundamental law-like first rules to decide ethical problems (dilemmas).



The Trolley Problem

The Scenario

A runaway trolley hurtles down a track towards five people who will be killed if you do nothing.

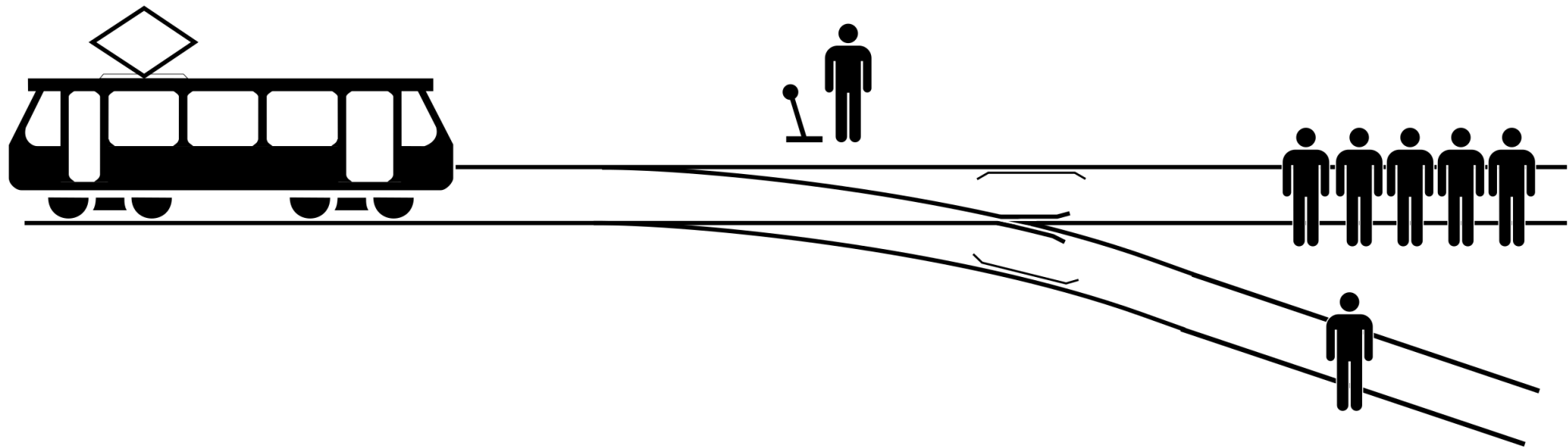
Your Choice

You stand at a lever. If you pull it, you'll divert the trolley to a side track where it will kill one person instead of five.

The Question

Do you pull the lever?

The Trolley Problem



By Original: McGeddon Vector: Zapyon - This SVG diagram includes elements from this icon:, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=67107784>

The Trolley Problem

Utilitarianism

→ PULL THE LEVER

- Five deaths versus one death
- Minimise total harm
- Clear utilitarian calculation
- Save more lives

The answer appears obvious: aggregate welfare demands pulling the lever.

The Trolley Problem

Deontology

→ DON'T PULL

- Pulling actively kills (not merely letting die)
- Uses person as means to save others
- Violates duty not to kill
- Action itself wrong, regardless of outcome

Even if the outcome is "better," the act of killing violates fundamental duty.

The Trolley Problem

Virtue Ethics

→ IT DEPENDS

- What would a virtuous person do?
- Is pulling courageous or cowardly?
- Does it show care or callousness?
- Requires practical wisdom (phronesis)

No formula provides the answer, only developed character and practical wisdom.

Principlism

Rather than choosing one ethical framework, **principlism** identifies widely-shared ethical principles that can guide practical decision-making across different philosophical traditions.

Core Idea

Focus on mid-level principles that bridge philosophical theory and practical action

Origins in Biomedical Ethics

Developed by Beauchamp & Childress (1979) for medical ethics, now widely applied to AI and technology ethics

Practical Advantage

Actionable guidance without requiring commitment to a single philosophical framework, useful for multidisciplinary teams

Key Principles in AI Ethics

Fairness

Treat individuals and groups equitably; avoid discrimination and unjust bias

Transparency

Make AI systems understandable and explainable to appropriate stakeholders

Accountability

Clear assignment of responsibility for AI system outcomes and impacts

Privacy

Protect personal data and respect individuals' informational self-determination

Safety & Security

Ensure systems are robust, reliable, and secure against misuse or failure

Human Autonomy

Preserve human agency, choice, and meaningful control over AI systems

These principles often conflict in practice: fairness vs. accuracy, privacy vs. transparency, autonomy vs. safety. Trade-offs must be taken. Principles found across EU AI Act, UNESCO recommendations, IEEE standards, and major AI ethics guidelines.

When Principles Conflict

Principles rarely align...

Common Trade-offs in AI Systems

Fairness ↔ Accuracy

Improving fairness across groups often reduces overall accuracy. Which matters more? For whom?

Privacy ↔ Transparency

Explaining how an AI works may reveal sensitive data. How much transparency can you provide without compromising privacy?

Autonomy ↔ Safety

Preventing harmful choices restricts user autonomy. Should the system allow errors to preserve freedom?

Efficiency ↔ Accountability

Full human oversight slows decisions. How much automation can you introduce while maintaining responsibility?

Designing AI Systems

For example: Fairness

The Principle

Core idea: AI systems should treat people equitably. No discrimination based on protected characteristics

Common definitions:

- Demographic parity: Equal rates across groups
- Equalized odds: Equal error rates
- Individual fairness: Similar individuals treated similarly

The Problem

Mathematical Impossibility Theorem

It is impossible to simultaneously satisfy all fairness criteria

What this means: Fairness is not a technical problem to solve but a **political choice** you must make

You must decide: Which definition? For whom? At what cost?

Design Always Encodes Values

Hostile Architecture



Public benches with armrests in the middle

Purpose: Prevent sleeping/lying down

Target: Homeless people

Design choice: Whose comfort matters?

Spikes on ledges and flat surfaces

Purpose: Prevent sitting or sleeping

Target: Homeless people, skateboarders, youth

Design choice: Whose use is "legitimate"?

Uneven "Camden benches"

Purpose: Can't sleep, skateboard, or hide things under

Label: "Defensive urban design"

Design choice: Who belongs in publicspace?

Design is never neutral. Every choice privileges some uses and users over others.

AI amplifies these choices.

Distributed Responsibility

The Distribution of Agency in AI Systems

Agency in artificial intelligence systems operates across multiple interconnected layers, each with distinct decision-making power and responsibility deflection mechanisms.

Developers

- Choose algorithms and architectures
- Select training data
- Define metrics and objectives
- Implement guardrails (or don't)

"I just optimised for accuracy"

Companies

- Decide what problems to solve
- Choose deployment contexts
- Set policies and monitoring
- Determine resources allocated

"We're just meeting market demand"

Users

- Decide whether and how to use
- Interpret and apply outputs
- Verify (or don't verify) results
- Provide feedback

"I just used what the system recommended"

Regulators & Society

- Set requirements and standards
- Enforce (or fail to enforce)
- Allocate liability
- Shape incentives

"We're still figuring out the rules"

When Agency Attribution Goes Wrong

When we attribute agency to AI, human responsibility disappears

Hiding Behind "The AI"

"The AI decided not to hire you"

"The algorithm determined your creditworthiness"

"The system flagged your content"

Reality: Humans designed the system, chose the data, defined the objective, deployed it, and continue to use it

Responsibility Disappears

Users say: "The AI told me to do it"

Designers say: "I just built what was requested"

Companies say: "The system made an error"

Regulators say: "It's too complex to understand"

Result: No one takes responsibility. Harm occurs without accountability.

Three Ways Responsibility Disappears

Many Hands Problem

When everyone is partially responsible, no one accepts full responsibility. "Not my job" — everyone points to someone else, creating diffusion of responsibility. This mirrors the classic Kitty Genovese bystander effect, now manifested in technological systems.

"Computer Says No" Phenomenon

Decisions are attributed to "the system" rather than human judgement. This creates psychological distance from outcomes and obscures the human choices that built and configured the system. "The algorithm decided, not me."

The Complexity Excuse

Systems become too complex for anyone to fully understand. "We can't explain how it works" transforms opacity into a shield against accountability. "It's a black box—even we don't know why."

❏ These are not inevitable technical limitations. These represent human choices about how we structure systems and accountability frameworks. We could design differently, maintaining clear accountability – but this requires deliberate, sustained effort.

Attributing Agency to AI

Why We Attribute Agency to AI

Humans naturally attribute agency to systems that exhibit certain characteristics, even when those systems have no real understanding, intentions, or consciousness.

Language Fluency

When systems produce coherent, contextually appropriate language, we instinctively treat them as understanding. Our social cognition is triggered by linguistic competence, even when it's purely statistical pattern matching

Anthropomorphic Design

Chatbots with names, first-person language ("I think," "I believe"), avatars, and human-like responses encourage us to treat AI as an agent with mental states. This is often intentional design, not accident.

Opacity & Complexity

When we can't understand how a system works, we fill the gap by attributing it agency ("the algorithm decided," "the AI wants") rather than acknowledging our own ignorance about the mechanism

The ELIZA Effect (1966)

What ELIZA Was

A simple chatbot created by Joseph Weizenbaum at MIT that simulated a psychotherapist using basic pattern matching and scripted responses

Example exchange:

User: "I'm worried about my mother"

ELIZA: "Tell me more about your family"

The mechanism: Simple keyword detection + pre-written response templates. No understanding whatsoever.

What Happened

Users attributed deep understanding and empathy to ELIZA – even when told it was just a program

Weizenbaum's secretary asked him to leave the room so she could have a "private" conversation with ELIZA

Some psychiatrists proposed ELIZA could automate therapy

Surface-level linguistic competence triggers attribution of agency, understanding, and care

Dr. Nim



AI Agency

Instrumental vs. Genuine Agency

A critical distinction for understanding AI ethics:

Instrumental Agency

"Acting as if it has purposes"

- Appears goal-directed
- Optimizes for objectives
- Adapts behavior
- Produces coherent outputs
- Described "as if" intentional

✓ AI systems have this

Genuine Agency

"Actually having purposes"

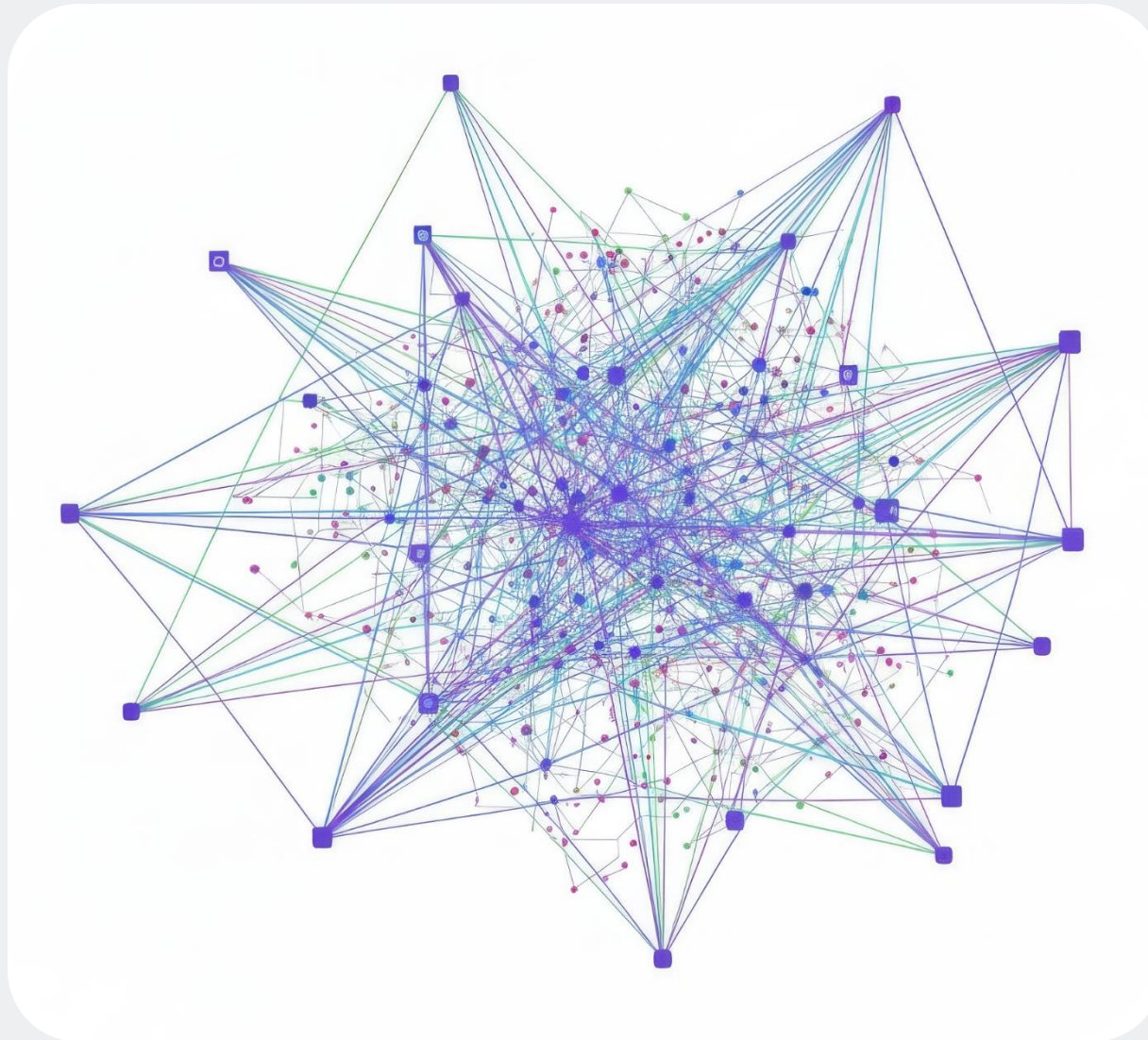
- Self-determined goals
- Conscious intentionality
- Understanding of stakes
- Moral responsibility
- Autonomy and freedom

✗ AI systems (currently) lack this

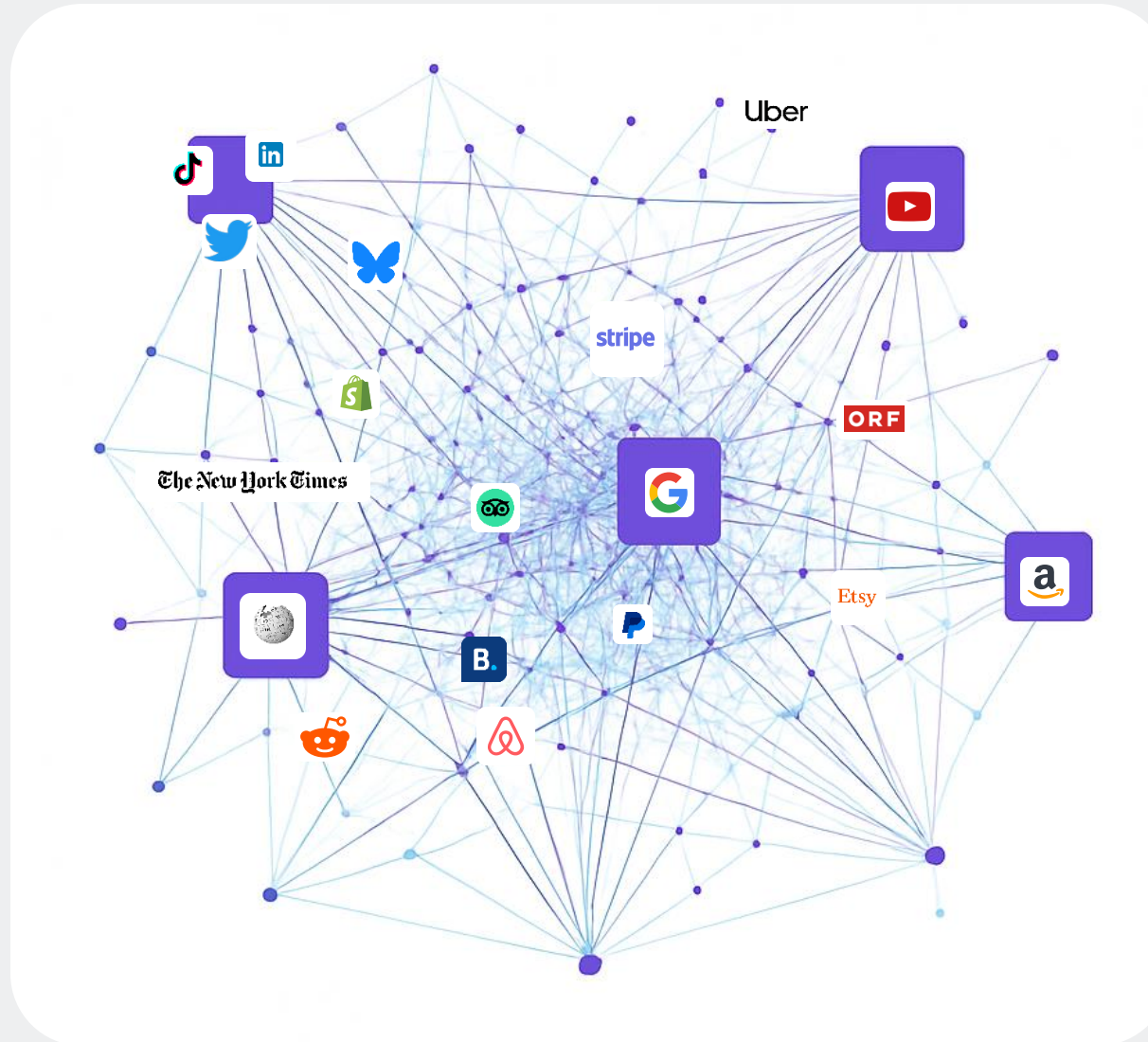
Using AI Systems

Inter-Human Relations

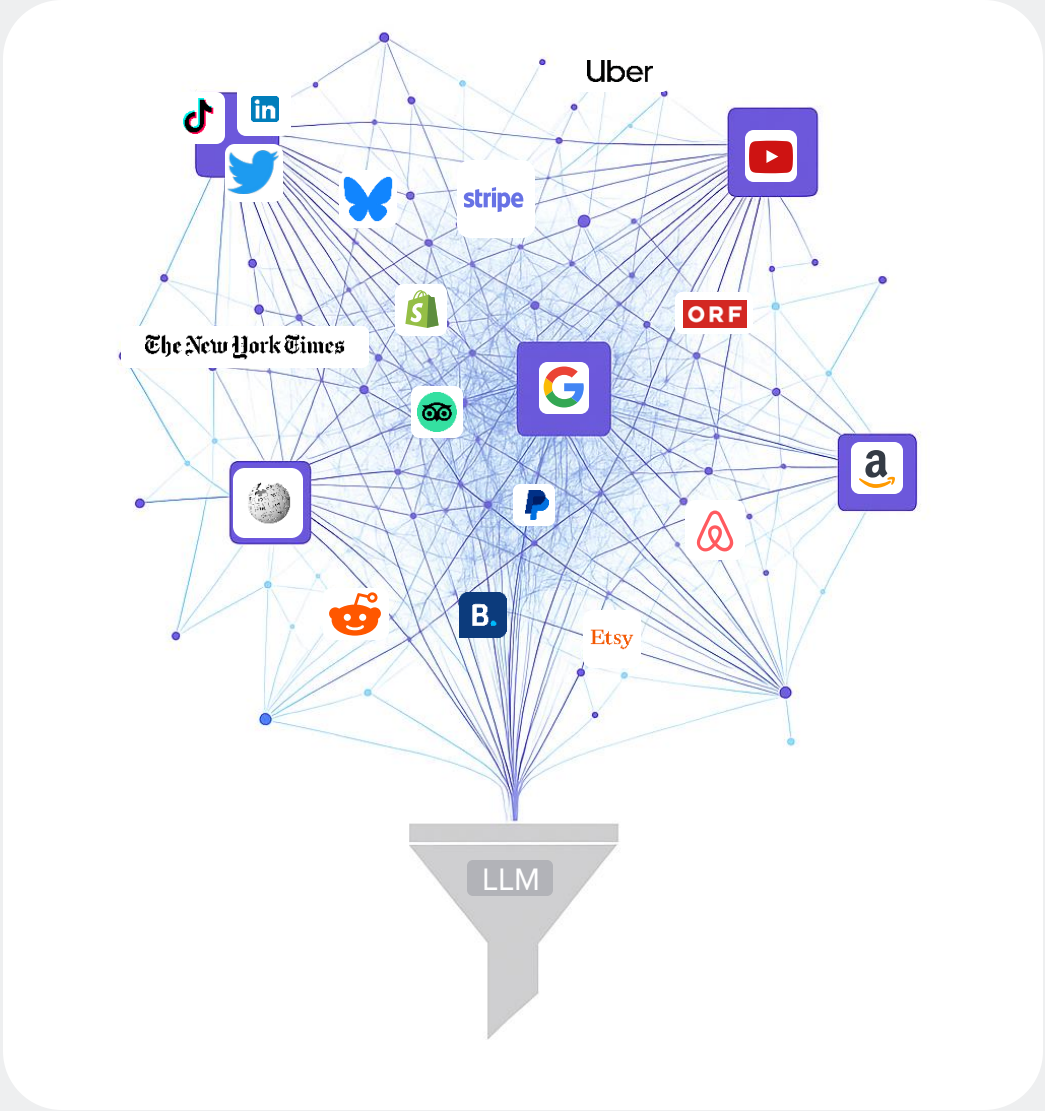
The Web

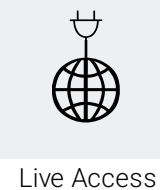
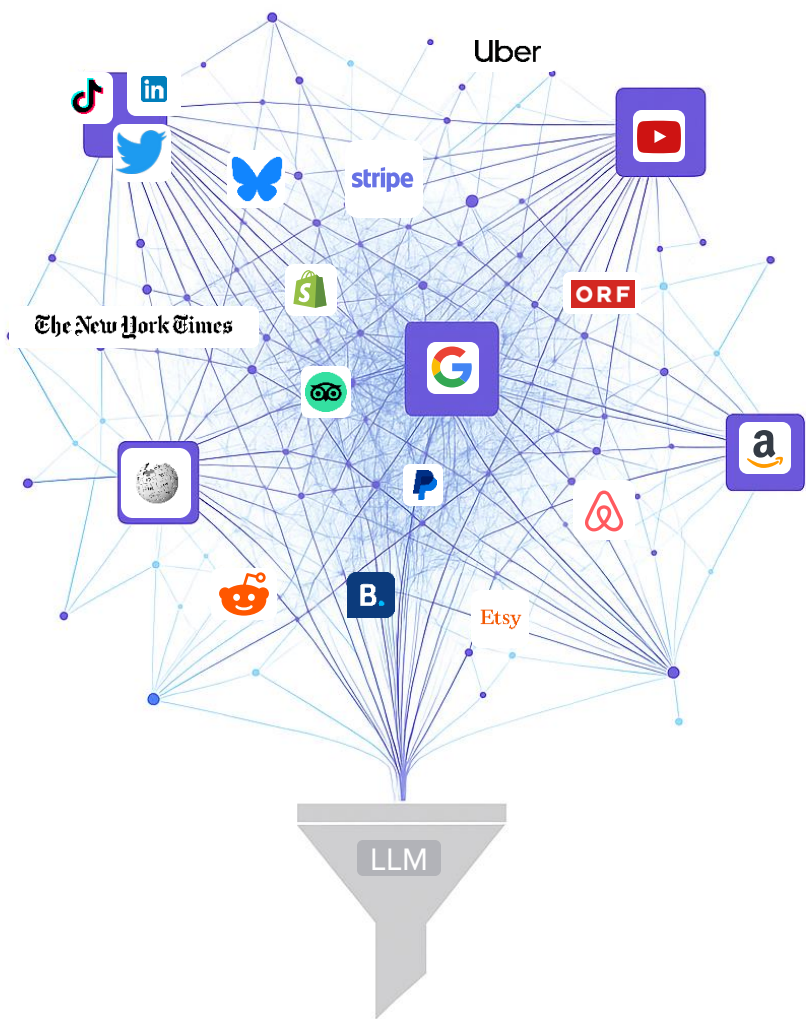


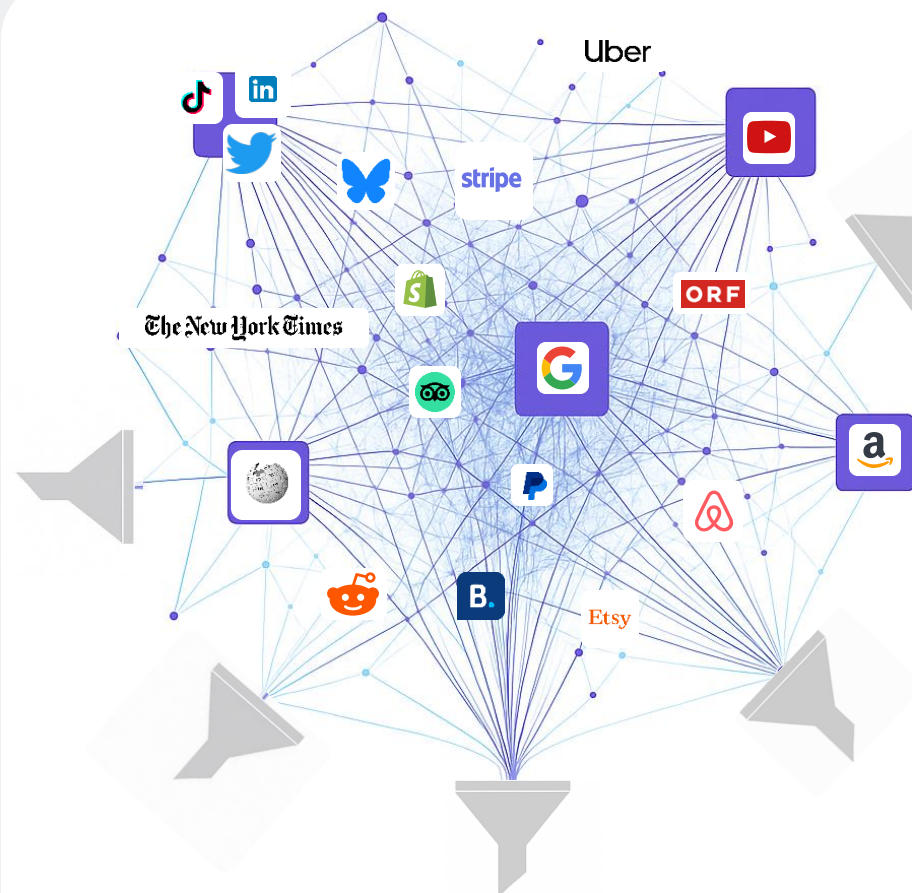
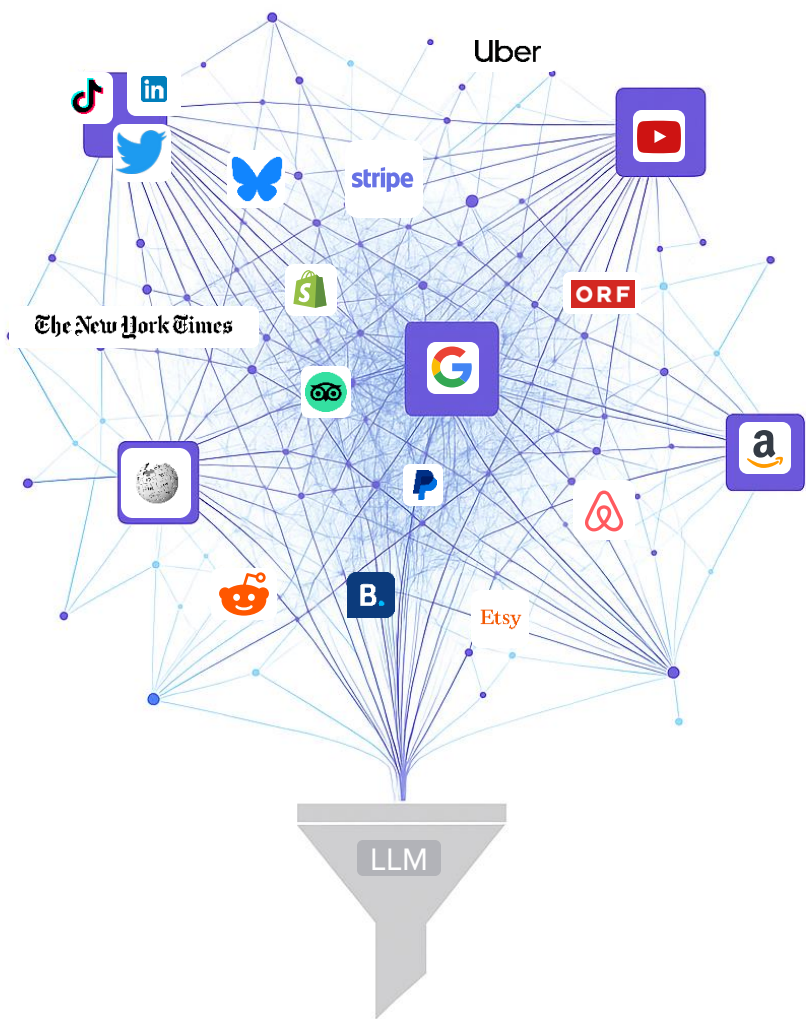
Until recently



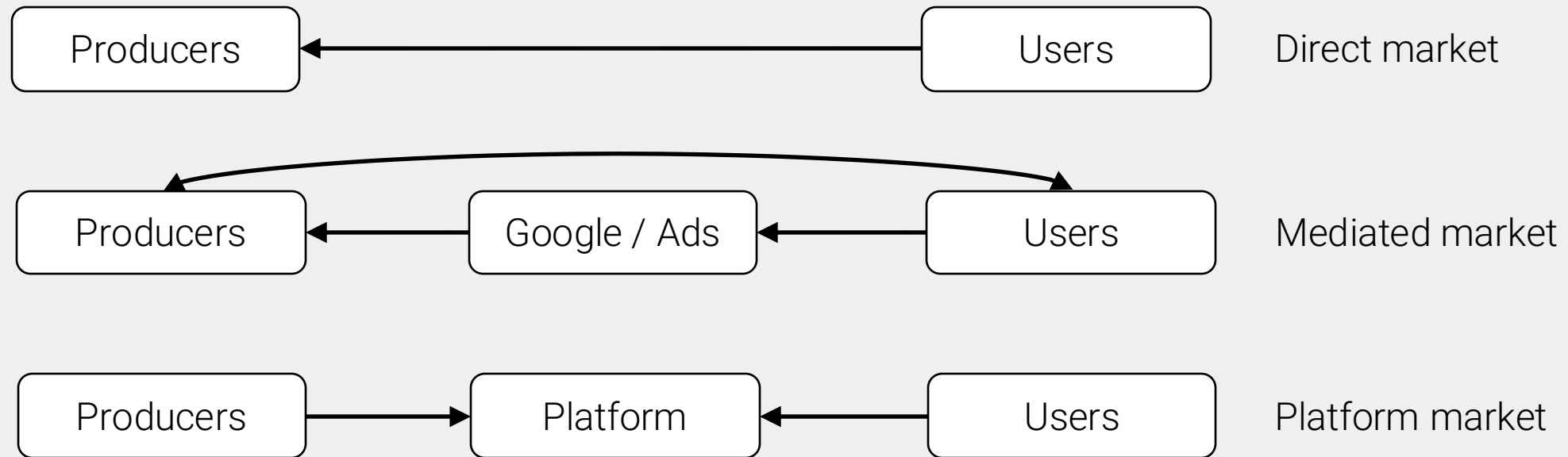
Now



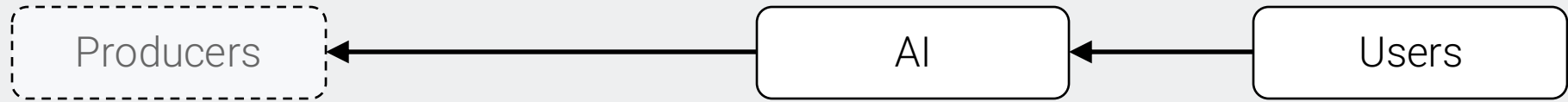




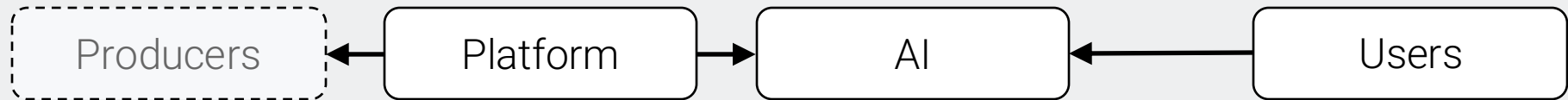
Market structure



Now



...or maybe



And then...

Fakes

Disruption

Intelligence as a Commodity

For the first time in history, cognitive capabilities can be purchased at scale like electricity or computing power

What This Means

Pay for Intelligence, Not People

Companies can purchase cognitive capabilities (writing, analysis, coding, translation) without hiring humans—API calls instead of salaries

Democratization and Concentration

Anyone with internet access can use sophisticated AI (democratizing)—but a few companies control the infrastructure (concentrating power)

Economics Fundamentally Shift

Marginal cost of cognitive work approaches zero—one AI model serves millions. Creates winner-take-all dynamics and new forms of inequality

Two Industrial Revolutions

First: Muscle Power on Demand

18th-19th century: Steam engines, machines, factories

What changed: Physical labor became available as a commodity—you could buy muscle power

Impact on work: Manual labor displaced, new jobs emerged, massive economic restructuring

Power shifted: Those who owned factories and machines accumulated wealth

Yet today: People still exercise, play sports, enjoy physical activity—even though machines are stronger

Second: Intelligence on Demand

21st century: LLMs, AI systems, automation platforms

What's changing: Cognitive labor becoming available as a commodity—you can buy intelligence

Impact on work: Knowledge work displaced, new jobs emerging, massive economic restructuring

Power shifting: Those who own AI platforms and compute are accumulating wealth

The future: People will still think, solve problems, enjoy intellectual challenges—even though AI is faster

Impact on Jobs and Work

Most Vulnerable: Routine Cognitive Work

Examples: Data entry, basic coding, document summarization, translation, customer service scripts, basic legal research, initial medical diagnoses

Why: Pattern-based, large training data available, clear success metrics

Augmented: Complex Knowledge Work

Examples: Software engineering, scientific research, strategic planning, creative design, teaching, journalism

Why: AI assists but human judgment, creativity, and contextual understanding remain essential. **Will this lead to augmentation or deskilling?**

More Protected: Physical & Social Work

Examples: Healthcare (hands-on), trades (plumbing, electrical), childcare, eldercare, therapy, arts performance

Why: Requires physical presence, human connection, trust, or real-world manipulation that AI cannot (yet) replicate

Power Concentration

AI amplifies existing power dynamics. Those with resources gain disproportionately

Compute & Infrastructure

Training frontier models requires billions in investment. Only a handful of companies can afford this scale.

Data & Network Effects

More users → more data → better models → more users. Winner-take-most dynamics.

Productivity Gains

Workers become more productive, but who captures the value? Historically: capital owners.

What Endures?

AI Will Do Everything

"AI will replace all knowledge workers" • "Human intelligence is obsolete" • "Nobody will write code anymore"

Human judgement and control

Some tasks automated, others augmented. Many require human judgment. The question is not "will AI replace you" but "how will you work with AI?"

The Sports Analogy: Using Your Mind Despite AI

Even though machines are far stronger, people still run marathons, climb mountains, do crafts by hand, and value physical fitness.

Similarly: People will continue to think, learn, and create—not because it's most efficient, but because *using our intelligence is intrinsically valuable*.

The Sports Analogy

Physical Activity

Machines are **far stronger** than humans, yet people still:

- Play sports for enjoyment
- Exercise for health and challenge
- Value physical accomplishment
- Find meaning in using their bodies

Cognitive Activity

AI is **far faster** than humans, yet people will still:

- Think for enjoyment
- Solve problems for satisfaction
- Value intellectual accomplishment
- Find meaning in using their minds

Just because AI can do something faster doesn't mean humans won't choose to do it themselves

Conclusion